THE UNIVERSITY OF MELBOURNE

GEOM90007 ASSIGNMENT 3 BRIEFING



INFORMATION VISUALISATION

YOUR OBJECTIVE

To work in a **team of four**
to create a **rich, interactive data visualisation interface**
targeted at a specific **user group**
and present your work to a **non-technical client**.

Assignment 1
Data visualisation
Limited interactivity

Assignment 2
Data visualisation
Interactivity
Sourcing data
Coding
Design summary

Assignment 3
Data visualisation
Interaction design
Sourcing data
Coding
Design summary
Audience use cases
Video pitch
…as a team of 4

THE TASK

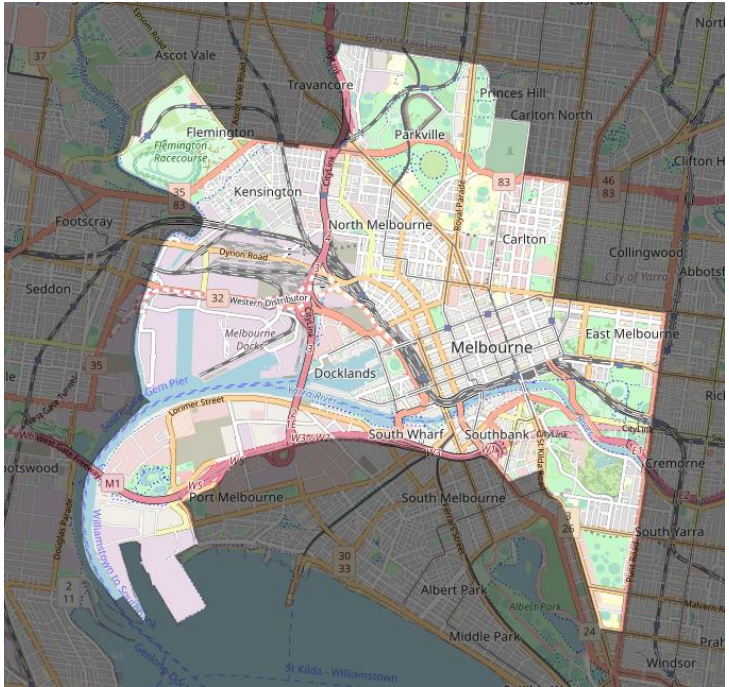
Your client has asked you to produce an **interactive interface** targeted at a selected user group.

Option 1: Tourism	**Option 2: Government performance**	**Option 3: Own choice (approval required)**
Client: The City of Melbourne *User group:* Tourists who are planning to visit Melbourne **or** are currently visiting Melbourne	*Client:* One of the ten departments of the Victorian Government *User group:* Public servants **or** members of the general public interested in the performance of any State Government-related topic (transport, planning and building, health, education, ...)	*Client:* As Option 1 or Option 2 *User group:* If you would like to target a different user group, please discuss with the Senior Tutor, Alan Thomas, and get approval from him.

THE TASK

Option 1: **Tourism**

- The **City of Melbourne** (your client) has asked you to produce an **interactive interface** to help tourists discover **interesting and practical information** about Melbourne.
- You may target
  - tourists who are planning to visit Melbourne **or**
  - tourists who are currently visiting Melbourne



The boundaries of the City of Melbourne – your interface should largely focus within these boundaries (can include neighbouring areas when relevant)

THE TASK



Option 2: Government performance

» A department of the **Victorian Government** (your client) has asked you to produce an **interactive interface** to expose **interesting patterns in data** and/or empower **decision-making** for specific tasks

» You may target

- public servants **or**
- members of the general public interested in the performance of any State Government-related topic (transport, planning and building, health, education, ...)

THE TASK

Option 3: **Own choice**

- Client is the City of Melbourne or a Victorian Government department
- User group – your own choice
- Get approval from Alan



HOW TO COMPLETE THE TASK

- Work in a **group of 4**
- Prepare an **R Shiny interface** that incorporates one or more **Tableau visualisations**

- Think about:
  - **When and where** will the interface be used?
  - What places are **most popular**?
  - What **patterns** can be found? How can you help the user to discover patterns?
  - Are there any **anomalies or specifics**?
  - What **platform** should be used (computer, mobile, kiosk, …)?

DELIVERABLES – WHAT YOU NEED TO SUBMIT

Part A Interface and report (25%)

Interface implementation

- A single **.zip** file containing your interface and all required data

Report

- Design summary (2 pages)
  - How your interface supports your chosen audience and purpose
  - Aspects of the design you want to get credit for, and justification of design decisions
- Pattern or use case summary (1.5 pages)
  - Interesting patterns or info that your target audience can discover using your interface
  - How your tool helps to discover these
- Sources and references

DELIVERABLES – WHAT YOU NEED TO SUBMIT

Part B Video (10%)

- 10 minutes
- Pre-recorded pitch presentation to your client (City of Melbourne or Victorian Government department)
- Provide background for, and demonstrate, your interface
- All team members must speak

Part C Peer evaluation

- Available after the final deadline
- Evaluate your teammates' contribution level

HOW TO SCORE WELL

What does a great submission look like?

- A very well-designed, revealing, logically structured, highly interactive, polished Shiny app containing Tableau viz(zes)
- It incorporates a fresh and innovative yet effective design, and demonstrates that you have challenged yourselves technically
- Must be designed firmly with the target user group in mind

Assessment criteria

- Basic design
- Technical challenge
- Design innovation
- Report

Note: your interface should include technical challenges beyond the lab sessions and Assignments 1 and 2.

WORKING TOGETHER

- Your group members will have different backgrounds
  - For example: uncomfortable with coding / may have a lot of experience with mapping / may not be strong at aesthetic design / …
- Discuss your individual skills and strengths
  - What are you each good at?
  - What would you not like to work on?
- Stay in contact
  - Agree on a weekly meeting time via Zoom (or other tool)
  - Reply to your group members' messages; do not "disappear"
- We will check that all team members have been contributing
  - If there are issues in your team, do not leave it until the end

RESOURCES

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Tableau

- Week 8 pre-lab videos on integrating Shiny and Tableau
- Publish to Tableau Public
  - Note: This makes your work publicly visible and searchable. Do not submit private or copyrighted datasets to Tableau Public.
- Actions in Tableau
- Tableau Embedding API v3
  - When embedding Tableau into Shiny, you can incorporate small pieces of JavaScript code into your R Shiny code using the **runjs** function. For example, you can make Tableau update a filter when a user clicks something on a Leaflet map in Shiny.
  - Tableau Embedding API v3 Tutorial

RESOURCES

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R and Shiny

» Training materials listed in Weeks 4-7

» R and Shiny textbooks listed in the Assignment 2 page on LMS

» R Graphics Cookbook

» ggplot2 cheat sheets | ggplot2 workshop video | ggplot2-book

» ggiraph-book

» Learn Shiny

» Leaflet Tutorials | R leaflet package

RESOURCES

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Web programming

Advanced students may wish to include web programming in their Shiny app.

- HTML for custom content: Using HTML in Shiny | HTML with W3Schools
- CSS for custom styling: Using CSS in Shiny | CSS with W3Schools
- JavaScript for custom interactivity: JavaScript with W3Schools | tutorial | guide
- Web programming cheat sheets:
  - HTML: This cheat sheet includes a list of every HTML tag, in alphabetical order.
  - CSS: There are many CSS properties, and it's easy to forget the specific name for one particular visual modification you want to make. This cheat sheet lists all of them.
  - JavaScript: This cheat sheet provides an overview of many basic aspects of JavaScript.

RESOURCES

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Visualisation tools

» Non-spatial visualisations
  – Tableau's many built-in visualisations (Lab 2) | R **ggplot2** and **ggiraph** (Labs 4 to 6)
  – 30 R Packages For Data Visualization That You May Not Know Of
  – Research other available R packages
  – Invent your own visualisation type! Draw a static visualisation using R graphics or (for advanced coders) make an interactive graphic using d3.js and the **r2d3** R package

» Geovisualisation
  – Tableau maps (Lab 3) | R **leaflet** (Lab 7)
  – Make your own custom basemaps using Mapbox Studio and add them to **leaflet** using the **mapboxapi** R package



RESOURCES

Selecting data

From LMS

Potential sources of data for your interface include:

- City of Melbourne Open Data Portal
  - Example dataset: Landmarks and places of interest
- Victorian Government open data
  - Example dataset: Open space
- Government Department websites
  - Example dataset: Rental Report
- A good data source directory can be found at the University of Melbourne Library: Australian GIS data guide

You are encouraged to start with exploring and analysing datasets in Tableau, Excel, or (in the case of geospatial data) GIS software such as QGIS. You will then have a choice to export your chosen datasets into other formats if required. These types of files can then be used in the finished interface.

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Thank you!

